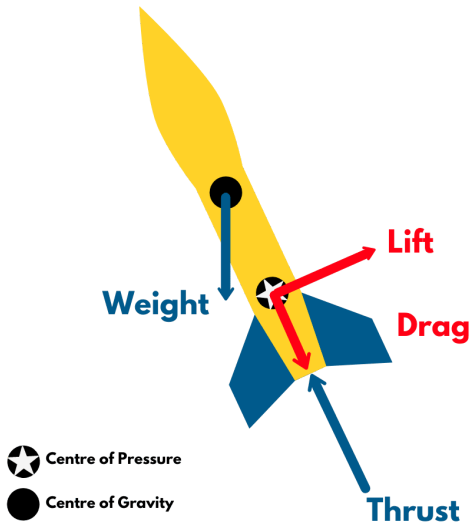
MAKING A ROCKET FROM SCRATCH

Step1: Designing Phase

The first step to any manufacturing process is starting with an insane design plan.

1. Aerodynamics of Rocket:

Let's start by understanding the basic aerodynamics of the rocket.



1.1 Forces on the rocket:

There are four key aerodynamic forces on a rocket:

1. Thrust (T): Generated by the rocket engine pushing gases out of the nozzle. Must overcome drag and gravity to accelerate upwards.

2. Drag (D): The air resistance acts opposite to the rocket's motion.Increases with velocity and depends on shape, surface roughness, and atmospheric density.
Parasitic Drag: Due to skin friction and shape.
Base Drag: Due to pressure differences at the rear of the rocket.

3. Lift (L): Lift is the aerodynamic force that acts perpendicular to the direction of airflow. It is generated when air flows over a curved surface

4. Gravity (W): Acts downward, pulling the rocket towards Earth.

https://www1.grc.nasa.gov/beginners-guide-to-aeronautics/four-rocket-forces/#external-factors

1.2 Rocket Stability:

Stability is the ability of a rocket to maintain its intended flight path without excessive deviation.

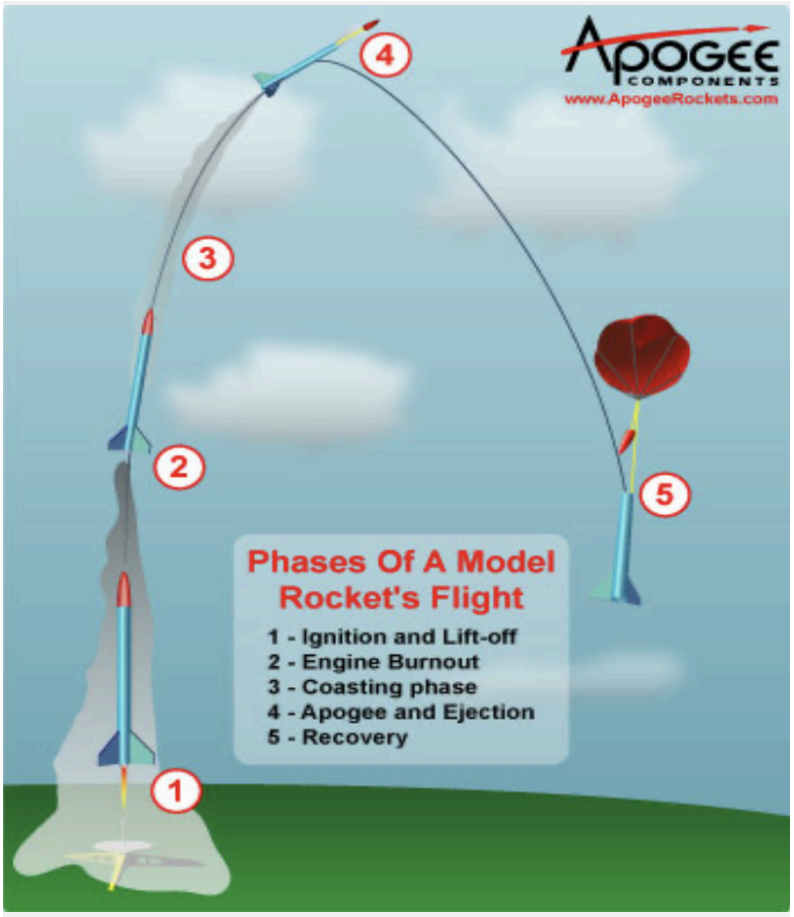
1. Center of Gravity (CG): The point where the entire weight of the rocket is balanced, and around which the rocket would rotate if external forces were applied.

2. Center of Pressure (CP): The point where the total aerodynamic forces (lift and drag) acting on the rocket are balanced. It is crucial for determining stability.

A stable rocket typically has its center of pressure behind the center of gravity.
Watch this video to understand why!

https://youtu.be/pnY5hkmJS58?si=SXHDvUlFnwCEJ2vg

1.3 Stages of a Rocket Flight:



1. Ignition and Liftoff

The rocket motor ignites, generating thrust to lift the rocket off the launch pad. It accelerates upward, overcoming gravity and atmospheric drag. The rocket is guided by fins or spin stabilization to maintain a stable trajectory.

2. Engine Burnout

The rocket burns all its fuel, and the engine stops producing thrust. Depending on the design, a single or multiple stages may burn out sequentially. At this point, the rocket's velocity determines how high it will travel.

3. Coasting Phase

With no thrust, the rocket continues upward due to inertia. It gradually slows down as gravity pulls it back. This phase lasts a few seconds to several minutes and is ideal for scientific experiments in near-microgravity conditions.

4. Apogee and Ejection

Apogee is the highest point in the flight, where the rocket momentarily has zero vertical velocity. If the mission involves deploying scientific instruments, small satellites, or parachutes, it happens at or just after apogee. Some rockets eject probes for atmospheric studies.

5. Recovery

The rocket and payload begin descending due to gravity. A parachute system may deploy to slow down the descent. The payload lands safely on the ground or in water, where recovery teams retrieve it for data analysis or reuse.

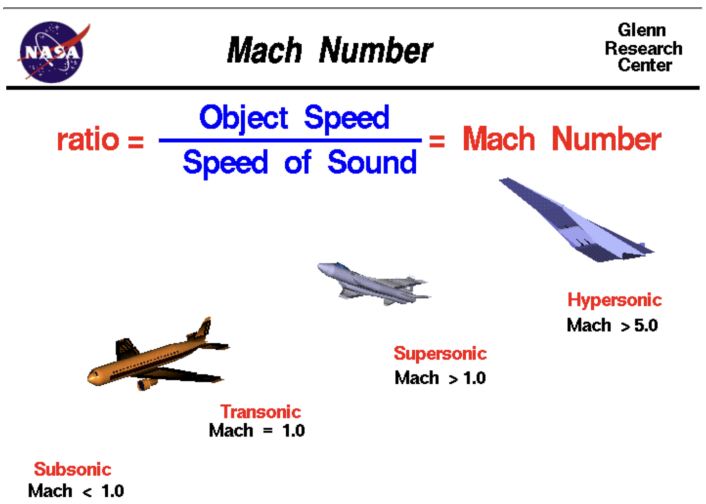
Descent Rate: The speed at which a rocket falls back to the ground after reaching apogee, typically controlled by parachutes or other recovery mechanisms.

1.4 Flow specifications:

When designing a rocket, it is very important to understand the type of flow and the flow parameters at which your rocket will be flying in.

Some terms that define your rocket requirements:

1. **Mach Number:**



At IITK RASET, since we build sounding rockets, we generally deal with subsonic speeds.

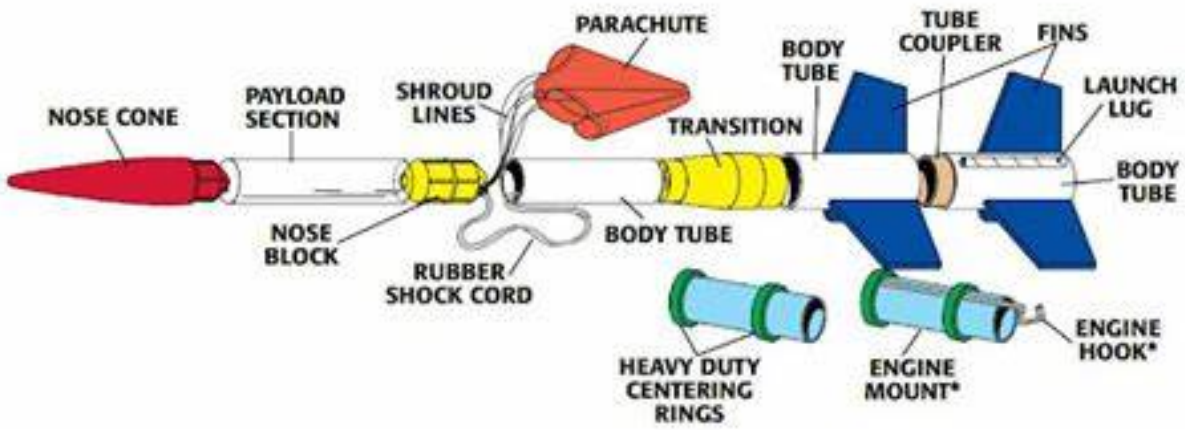
2. **Reynold's Number:** It is a dimensionless number that deals with the nature of flow of the air. It determines whether flow around the rocket is turbulent or laminar.

$$\mathrm{Re} = \frac{\rho V L}{\mu} = \frac{inertial}{viscous}$$

Watch this video to understand more.
Laminar flow, turbulence, and Reynolds number

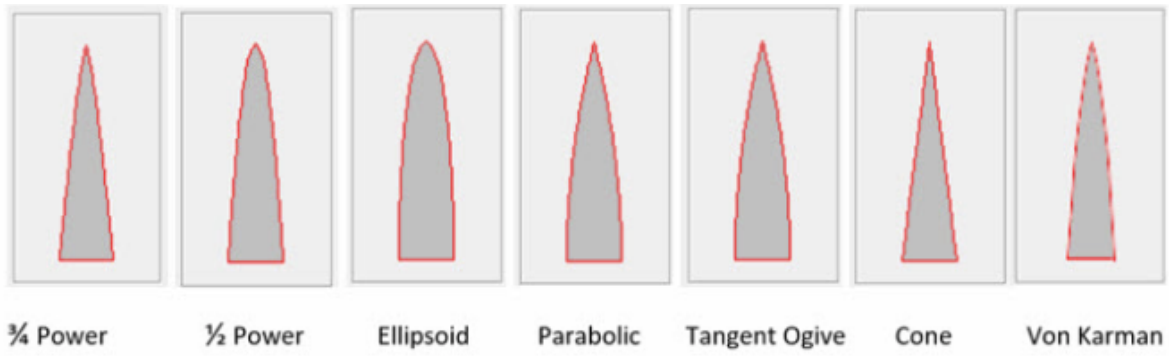
2. Components of a Rocket:

Now that we know about the aerodynamics aspects, let's move onto what the rocket actually consists of:



2.1 Nose cone:

A nose cone is the conically shaped forwardmost section of a rocket, guided missile or aircraft, designed to modulate oncoming airflow behaviors and minimize aerodynamic drag.



Read more about it here:

3/4 Power Nosecone: Defined by a power-law equation ($n = 0.75$), balancing elongation and volume efficiency. Offers moderate drag reduction and is used in applications requiring a compromise between speed and structural simplicity.

1/2 Power Nosecone: Blunter profile with a power-law exponent ($n = 0.5$), prioritizing payload volume over aerodynamics. Ideal for subsonic speeds where drag minimization is secondary to space utilization.

Ellipsoid Nose Cone: Smooth, rounded shape resembling a stretched sphere, optimized for subsonic aerodynamics. Commonly used in aircraft and low-speed rockets to minimize drag.

Parabolic Nose Cone: Quadratic-curved profile effective at managing shock waves in supersonic flows. Reduces wave drag and enhances stability for high-speed missiles and aerospace vehicles.

Tangent Ogive: Curved nose transitioning smoothly (tangent) to the body, minimizing supersonic drag. Widely employed in missiles and high-performance aircraft for aerodynamic efficiency.

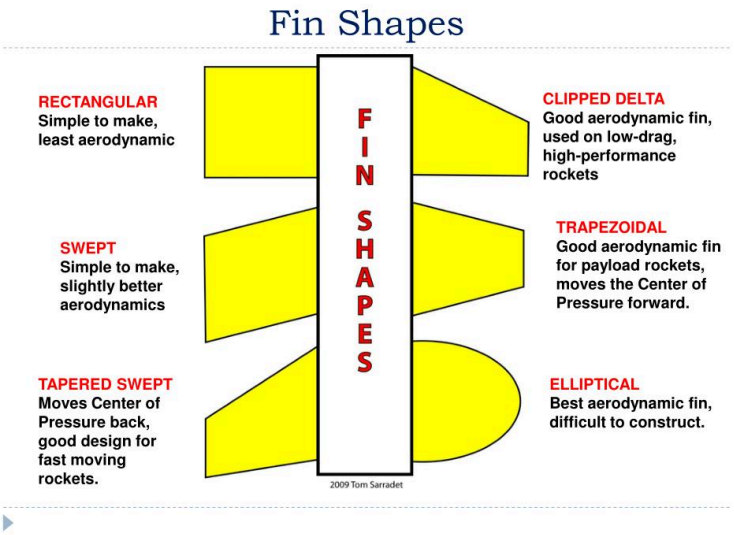
Cone Nose Cone: Simple conical design with straightforward manufacturing and adequate supersonic performance. Favored in cost-sensitive applications where advanced shaping is unnecessary.

Von Kármán Nose Cone: Mathematically optimized ogive shape for minimal supersonic wave drag. Preferred in high-speed aerospace systems like re-entry vehicles for peak aerodynamic efficiency.

[Richard Nakka's Experimental Rocketry Site]

2.2 Fins:

The reason rockets have fins has to do with stability. The fins help the rocket keep pointing in the direction it launched. When a rocket is flying through the air, changes in the air can make the rocket wobble. If it wobbles too much it can go off course. Having the right size, shape, and amount of fins will help make sure your rocket corrects itself when it wobbles. The size, shape, and number of fins will change things like your rocket’s center of mass, and how much drag it faces when it flies.



Understanding of fins:
[Sizing Fins | STAR Public]

Read more about it here:
[nakka-rocketry.net/RD_fin.htm]

Different parameters of Fins
[https://rocketry.gitbook.io/public/tutorials/airframe/sizing-fins]

1. Flutter: A dynamic aeroelastic instability that occurs when aerodynamic forces couple with structural vibrations, leading to oscillations that can grow uncontrollably and cause structural failure.

2. Divergence: A condition in which an aerodynamic surface (such as a rocket fin or wing) bends under aerodynamic loads, leading to an unstable increase in deflection and potentially causing structural failure.

2.3 Body Tube:

These are cylindrical shaped tubes that forms the main body of the rocket. It typically houses the rocket's payload, propulsion systems, and other necessary components. It is generally made up of light-weight material whose strength-to-weight ratio allows the rocket to remain sturdy while minimizing overall weight.
The shape and size of the body tube play a crucial role in the rocket's aerodynamics. A streamlined design helps reduce air resistance, allowing the rocket to travel more efficiently through the atmosphere.
The body tube often contains a compartment where the payload, avionics and recovery are housed, with the help of bulkheads and centered rings.
Joining of two body tubes can be done using a coupler.

Read more about it here:
[Richard Nakka's Experimental Rocketry Site]

2.4 Centered Rings:

Centering rings stabilize internal components like the motor, payload, avionics, and recovery system by keeping them securely aligned within the airframe. They prevent shifting, distribute loads, and enhance structural rigidity, ensuring a stable flight and safe recovery.

2.5 Bulkhead:

A bulkhead is a structural plate inside the rocket that separates different compartments (e.g., payload bay, avionics bay, or motor section) and provides mounting points for components. It helps with structural integrity, pressure sealing, and recovery system attachment.

2.6 Coupler:

A body tube coupler is a structural component used in model and high-power rockets to connect two sections of the rocket's body tube. It is typically a short cylindrical piece made of materials like cardboard, fiberglass, or carbon fiber, designed to fit snugly inside the body tube sections it joins. Couplers provide additional strength and stability to the rocket's airframe while also allowing for modular assembly, making transportation and storage easier. In some designs, they may house electronics, recovery systems, or serve as separation points for multi-stage rockets or deployment mechanisms.

3. Choice of Material:

When building a rocket, choosing the right material is of utmost importance.

Go though the link to find some common materials used:
https://www.rs-online.com/designspark/material-choices-for-model-rocketry

Also, try finding the differences between carbon fibre and fiberglass and look up their advantages and disadvantages.

Note:

We have outlined some of the basic concepts of Rocketry and aerodynamics for you but please go through the accompanying links to properly understand the concepts.
P.S. We will be testing you on your enthusiasm for rocketry so extra reading always goes a long way ;)

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